

News architecture V2: audit, implementation and growth plan

Decision date: 9 September 2026. **Scope:** the existing MVP, its separate news service, 1,515 catalogue rows, and the path to 10,000–50,000 companies.

Decision

Keep the saved V1 as a reference and use **Global News 2** for the next news experience. V1 demonstrates aggregation and company feeds, but its stored coverage, source classifications and polling behaviour do not establish dependable current coverage. V2 adds a free official-release feed, an internal article reader, explicit optional provider connections, company and portfolio filtering, and visible source health.

Free users can read supported official releases without subscribing to a publisher. This does **not** provide every publisher’s full articles, comprehensive international company news, or guaranteed real-time coverage. A complete global commercial news product will need contracted coverage. Benzinga is the first full-text integration candidate; Marketaux is a candidate for global entity-tagged **snippets**. LSEG/Reuters and Factiva deserve enterprise evaluation when their coverage and display licences justify the cost. [1–11]

The working implementation is a **local single-workspace MVP**. Multi-tenant authentication, contractual entitlement enforcement, production retention rules and scale benchmarks remain release requirements. Adding 50,000 CSV rows alone would not meet those requirements.

Saved version and rollback

Before changing the architecture, both repositories were committed and tagged `snapshot/news-v1-polymarket-20260909` :

Repository	Baseline commit
MVP	6e6c2730ca2923c2563be7bb7bb8054e1b457aeb
Standalone news service	c909cc2d716b44d1b75967f4f57554321849e428

The private local backup directory is `.local-backups/news-v1-polymarket-20260909/` in the MVP. It contains a full workspace archive, consistent SQLite and PostgreSQL backups, verified Git bundles, a checksum manifest and `RESTORE.md`. Source, existing reports, screenshots, the MVP and Polymarket integration are included. Private configuration is included in the protected backup; it must not be uploaded to the public repositories. Regenerable dependencies and build caches are excluded. V2 work is on `feature/news-v2` in both repositories.

What the audit actually found

The MVP has **two news stores and two company universes**. `/news`, holdings and the original catalogue use SQLite and a 50-row CSV. `/global` uses the separate PostgreSQL service and the production CSV with 1,515 rows. Only 28 exact ticker strings overlap. The production file has 1,441 distinct exact company names; share classes and multiple listings must not automatically be treated as separate issuers. Another file contains 30,992 US instruments, including funds and other instruments; it is not evidence of 30,992 covered companies.

The live V1 database contained 38,244 articles and 52,423 company links. Of 1,515 active catalogue rows, 1,512 were resolved and 1,450 had some stored news. The latest ingestion run finished on **31 August 2026 at 17:31 UTC**, processed one company, and was roughly 202 hours old at the audit. Those numbers describe historical stored coverage, not a fresh complete feed.

Finding	User impact	V2 response / remaining work
Scheduler and runner acquired the same advisory lock on different connections	Scheduled ingestion could skip itself as “already running”	Remove the duplicate outer lock; keep exclusive execution in the runner
Non-ASCII characters were removed from the content hash	Different Cyrillic/CJK headlines could collapse into one	Preserve Unicode letters and digits; regression coverage
Timeout wrappers could leave network and limiter work running	Work continued after apparent failure; rate limits became misleading	Propagate cancellation; share redirect concurrency across companies
A paginated provider window could be marked complete after its first page	Missing stories became invisible to later incremental runs	Resume incomplete windows; never advance the stable watermark until completion
Provider names were used as a proxy for official-source quality	A syndicated report, issuer claim and official disclosure looked equivalent	Separate official, wire, aggregator and archive provenance; retain the original source
Any historical article could imply a successful connection	Old feeds looked healthy	Show per-provider last attempt, last success, errors and request budget
Existing company identity depended heavily on ticker strings	Cross-exchange collisions and false portfolio matches	Resolve using listing identity plus company evidence; show ambiguous/unmatched cases
Every headline opened a new tab	Users left the product immediately	Internal reader for all existing news entry points; original publisher remains an explicit option
General-feed image enrichment crawled publisher pages on a read	Slow reads, extra network requests and uncertain image rights	Use supplied images only; independent provider failures and a five-minute general-feed cache
Consumer/API credentials and display permission were not modelled separately	Technical access could be mistaken for a licence	Explicit display/full-text settings; paid connectors disabled by default

Some V1 defects are fixed to protect the comparison experience. V1 archive content retains its original, unverified permission status; moving it into a different UI does not grant new rights. The remaining production work includes authenticated access to all older routes, removal of the old Finnhub-key exposure endpoint, replacement of arbitrary-URL image enrichment still used by older paths, and tenant-aware storage. V2 does not use those mechanisms.

Architecture and data flow



Collection is source-driven: fetch a provider’s incremental stream once and associate each item with the relevant companies. Reading the directory or filtering a portfolio does not launch one external request per company. PostgreSQL stores jobs and checkpoints; a worker claims jobs with leases, refreshes leases, persists bounded batches and resumes unfinished windows. Source HTTP requests consume a shared database budget, including additional release-body and identity-mapping requests.

Each record includes the provider item ID, headline, supplied summary, permitted plain-text body, original URL, publisher, language, publication/update/receipt timestamps, topics, content mode, usage metadata, revision and deletion state. Company associations retain the evidence used to resolve them. Raw HTML is not rendered. Unknown image permission means no article image is shown; a neutral source mark is preferable to an invented news photograph.

Different providers' copies remain attributable records. A future cross-provider story cluster should group related records without discarding independent reporting, corrections or conflicting claims. The MVP does not claim to have implemented semantic event clustering or an editorial fact-checking system.

Sources, access and payment

Source	What the connector supplies	Free / permission position	Current implementation
Federal Reserve Board	Institutional policy/regulatory releases and selected release text	No API key or publisher subscription. Board information is generally reusable unless marked otherwise; retain attribution and exclude third-party materials/seals. [1]	Live free source
European Central Bank	Selected institutional press releases and text	No subscription. Accurate attribution and modification disclosure apply; paid products must disclose the original is available free. Authored papers/speeches are excluded from this connector. [2]	Live free source
SEC EDGAR	Filing metadata, forms, dates, CIK-linked disclosures; supported document text where obtained	Public access without an API key; honest User-Agent and SEC fair-access rules. These are disclosures, not a complete journalism feed. [3–4]	Implemented; live requests returned HTTP 403 here
Benzinga	Incremental financial news, company tags, updates, removal events, optional full bodies	Provider API token and the appropriate business/customer-display agreement. Full text depends on the purchased product and entitlement. Project price is unknown. [5–6]	Connection UI and adapter; no paid credentials or live paid-feed test
Marketaux	Global headlines, descriptions/snippets, links, entity metadata	Free developer quota exists; published terms do not establish unrestricted business display. Obtain applicable commercial terms. No full article bodies, including paid plans. [7–9]	Connection UI and adapter; not connected
Finnhub	General headlines/summaries or a bounded selection of North American company-news queries	Personal plans cannot be used by a business even internally without written approval. Display/redistribution requires the appropriate rights. [10]	Connection UI and adapter; existing V1 key is not automatically reused
LSEG / Reuters	Enterprise global news and story APIs, identifiers and topics	Contracted product, source entitlements and customer-display/redistribution rights; a terminal login is insufficient	Enterprise evaluation only; no pretend connection [11]
Dow Jones / Factiva	Enterprise discovery, streams, archives and entitled article retrieval	Contracted source/content/display rights; consumer subscription is insufficient	Enterprise evaluation only [12]

Marketaux's advertised monthly API plans, checked on 9 September 2026: free **100 requests/day, 3 articles/request**; \$29 **2,500/day, 20/request**; \$49 **10,000/day, 50/request**; \$99 **25,000/day, 100/request**; \$199 **50,000/day, 100/request**. These prices describe API plans, not a confirmed customer redistribution licence. Its "200,000+ entities" claim is not a guarantee that our particular 50,000 companies have current news. [8]

The free official-source path has no publisher subscription charge. Hosting, engineering and maintenance still cost money. For paid content, an end user's normal website login cannot be converted into API redistribution rights. The implemented forms accept provider API credentials, record the user's declared display entitlement and separately enable full text where supported. A checked box does not verify a contract; it is a local setup control pending a production entitlement system.

Reading inside the platform

The primary click opens an internal reader. Three honest content modes are available:

1. **Full article/release:** permitted text was actually retrieved and is rendered as plain text with attribution.
2. **News brief:** only the provider's supplied summary is available; the UI says so.
3. **Source link:** the stored item has only a headline and source URL.

This avoids silently dropping users into a new browser tab. Opening the original publisher is optional and explicit; that publisher may still require payment. No paywall bypass, subscriber-cookie forwarding, arbitrary publisher scraper, embedded browser login or iframe is used. ECB business links must open its own full page, so copying a permitted release into our reader is distinct from framing the ECB website. [2]

Existing `/news` All News/My Stocks, company detail news, `/global`, and global company feeds use stable internal reader links. The original general-news brief cache is bounded to 2,000 items/30 days for this preview; that engineering limit is not itself a licence. Company catalogue pages also lead to the matching V2 company view, and the catalogue has an entry to the full 1,515-row news directory.

Relevance, accuracy and freshness

Official source means identifiable institutional origin; it is not a guarantee that every claim is complete, impartial or correct. An issuer release is the issuer's statement. A wire story is editorial reporting. An aggregator is a delivery mechanism. V2 preserves those distinctions instead of assigning universal credibility from a publisher-name regex.

Company-specific feeds require a verified association. Unmatched/ambiguous holdings are visible and are never silently expanded into the global feed. Direct company associations and general macro context remain distinct. A central-bank announcement can be useful to a portfolio without proving that it specifically concerns every holding. Provider entity tags remain evidence to check; they are not causal predictions about a share price.

The local portfolio resolver accepts up to 200 distinct holdings and reports an explicit limit for larger portfolios. This is separate from the paginated global company directory. Initial live matching identified only Apple. The final version resolves **all three current holdings** after adding two reviewed mappings: ASML/NASDAQ from ASML's own share-listings page, and Alphabet Class A/GOOGL from its Q2 2026 filing. The mappings retain source URLs, review dates and a one-year review expiry; incorrect venues, countries and Class C substitutions remain rejected. V1 catalogue/listing rows were not changed. Unreviewed holdings are not quietly matched by ticker alone. [13–14]

V2 polls official sources every **15 minutes** by default and reads the local cache in visible browser tabs every **60 seconds**. Connected paid sources default to a five-minute collection interval, subject to their configured budget. This is **periodic refresh, not a real-time wire or guaranteed 60-second source latency**. Unchanged official feeds avoid fetching all bodies repeatedly; new/changed feed records are fetched, and unchanged release bodies receive a daily correction rescan while they remain in the bounded feed. A correction without a feed metadata change can therefore take about a day plus queue/budget delay to appear. Feed snapshots are not exhaustive archives.

The UI shows publication time separately from receipt/check time, partial errors, missing source coverage and exhausted quotas. A source checked today can legitimately have no release today. Successful polling does not prove that the upstream source published everything or that a company had no news.

Production targets should be agreed after measurement: 99% of eligible received items processed within two minutes, source-specific lag alerts after two missed collection intervals, correction/removal propagation within the contracted feed's update interval, and separately sampled association precision. These are proposed acceptance targets, not measured MVP SLAs. A multilingual labelled evaluation set should include ambiguous tickers, subsidiaries, funds, share classes, renames, delistings and negative examples.

Scaling from 1,515 to 50,000 companies

At one request per company and 55 requests/minute, a complete sweep takes approximately 27 minutes for 1,500 companies, 3 hours for 10,000 and 15 hours for 50,000—even before retries or pagination. Repeating a per-company query every five minutes would require 432,000, 2.88 million or 14.4 million requests/day respectively. Increasing thread count cannot overcome those quotas.

Layer	Implemented now	Required before sustained 10k–50k operation
Identity	Existing company/listing directory; conservative matching; visible unresolved cases	Audited issuer/security/listing hierarchy, permanent IDs such as LEI/FIGI/ISIN/CIK where licensed, aliases and corporate-action history
Acquisition	Bounded incremental provider jobs; shared source budgets	Contracted feed capacity, bulk initial backfill, source-specific completeness monitoring and recovery SLAs
Storage	Indexed PostgreSQL articles, associations, revisions and jobs	Retention/partitioning by measured volume; archival storage where licensed; backup restore drills
Search	Bounded text search and cursor pagination	Full-text/trigram or dedicated search index, language-aware ranking and relevance evaluation
Distribution	One local workspace; server-only credentials	Authenticated tenants and users, licence/user/source cache keys, per-request entitlement checks, audited revocation and deletion
Performance	Directory pagination avoids a 50k quote/news request fanout	Load tests at expected article count and concurrent users; query plans, queue lag, p95/p99 latency and failure budgets
Product	All/macro/portfolio/company views and source selection	Regional coverage policy, translations where licensed, editorial handling of disputed/corrected stories

Do not replace the root 50-company CSV with the production file without adapting its schema and quote/profile loading. The two files have different fields, country formats, name conventions and exchange identifiers. The new news directory accesses the full service catalogue without launching quote requests for every row.

Implemented interface and evidence

The separate `/global-v2` page includes the feed, full directory, portfolio/company views, topic/source filters, search, pagination, provider settings, request status and an internal reader. Real-source screenshots and a machine-readable validation snapshot accompany this report under `screenshots/news-v2-*` and `news-v2-evidence/2026-09-09/`. Test fixtures, if used for failure-case UI tests, are not presented as live source evidence.

The final live check found **20 Fed releases, 6 ECB releases, zero SEC records**, and no connected commercial feed. All three sample holdings resolve, but **zero companies yet have newly ingested V2 company news**: the free records currently available are macro/regulatory releases, and SEC access is blocked. Matching a holding is not the same as supplying fresh company news. The separately selectable V1 archive supplies historical company briefs with unverified display rights and visible publication dates.

Validation completed: **127 offline news-service tests, 6 PostgreSQL integration tests with transaction rollback, 19 Polymarket regression tests, 14 read-only local API checks**, production builds and targeted TypeScript/ESLint checks. Browser tests covered real release-body fidelity, internal/reloadable readers, provider forms, archive/company navigation, 390px mobile width and dark mode. Fixture-only tests additionally exercised revoked access, partial outage, credential handling and pagination. No paid provider credentials were available, so paid-feed ingestion and contractual entitlement have not been verified live.

Live interface examples

🔍 Search for stocks, ETFs, and more

🔊

My Stocks

News

Catalogue

Global News

Global News 2

Polymarket

Alerts 27

My Strategy

Settings

🌙

👤

INTELLIGENCE WORKSPACE

Global News 2

Read the developments behind your investments. Official releases, company reporting and portfolio context, together in one place.

🔧 Manage sources

🔄 Fetch latest

● Cached feed checked 9 Sept 2026, 04:04 UTC

Feed checks every 60 seconds · All times UTC

YOUR NEWSROOM

Global brief

All news

My portfolio

Macro & policy

Companies

EXPLORE TOPICS

Monetary policy

Inflation

Regulation

Earnings

Markets

Read inside the platform. Every story retains its source and content availability. [View source connections →](#)

The global brief

Official announcements and releases from enabled public sources.

🔍 Search headlines and summaries

Search

Public sources

All topics

1 source reports a failed last fetch. Available stories remain visible. [Review source status](#)

The latest available story in this selection was published 4 Sept 2026, 15:00 UTC. This feed reflects available source results, not a real-time news guarantee.

Federal Reserve Board · 4 Sept

Federal Reserve Board announces termination of enforcement actions with United Texas Bank, Quontic Bank Acquisition Corp., and Quontic Bank Holdings Corp.

regulation

Full article →

Federal Reserve Board · 27 Aug

Federal Reserve Board issues enforcement action with former employee of Banco Popular de Puerto Rico

regulation

Full article →

Federal Reserve Board · 25 Aug

Minutes of the Board's discount rate meetings on July 20 and July 29, 2026

monetary policy

Full article →

YOUR NEWS LIBRARY

Catalogue records 1,515

With V2 news 0

Enabled sources 3

Catalogue size measures available company records, not complete news coverage. [Explore the company directory →](#)

Source connections Manage

Federal Reserve Board

Public source

Last fetch succeeded

European Central Bank

Public source

Last fetch succeeded

SEC EDGAR disclosures

Public source

Last fetch failed

Benzinga

Your API connection

Needs setup

Marketaux

Your API connection

Needs setup

Finnhub

Your API connection

Needs setup

Freshness & coverage

Latest receipt

Captured from the running MVP on 9 September 2026. Publication dates, failed-source status and the distinction between catalogue size and news coverage are visible. Source illustrations are neutral interface artwork, not news photographs.

Search for stocks, ETFs, and more

My StocksNewsCatalogueGlobal NewsGlobal News 2PolymarketAlerts27My StrategySettings

YOUR NEWSROOM

Global brief

All newsMy portfolioMacro & policyCompanies

EXPLORE TOPICS

Monetary policyInflationRegulationEarningsMarkets

Read inside the platform. Every story retains its source and content availability. View source connections ->

Back to your feed

Federal Reserve Board · 4 Sept 2026, 15:00 UTCFull article

Federal Reserve Board announces termination of enforcement actions with United Texas Bank, Quontic Bank Acquisition Corp., and Quontic Bank Holdings Corp.

Version 2.

The Federal Reserve Board on Friday announced the termination of the enforcement actions listed below:

United Texas Bank, Dallas, Texas

Cease and Desist Order dated August 29, 2024 (PDF)

Terminated September 2, 2026

Quontic Bank Acquisition Corp., Astoria, New York, and Quontic Bank Holdings Corp., Astoria, New York

Written Agreement dated July 5, 2023 (PDF)

Terminated September 2, 2026

Additional enforcement actions can be searched for here .

For media inquiries, please email or call 202-452-2955.

Source: Federal Reserve Board. Board information is public domain unless otherwise indicated; third-party materials and seals are excluded. https://www.federalreserve.gov/disclaimer.htm
Public-source material; original attribution applies.
Received 9 Sept 2026, 03:58 UTC · Language: en

Open original source ↗Return to feed

Explore article topics

regulation ->

Your news library

Catalogue records1,515

With V2 news0

Enabled sources3

Catalogue size measures available company records, not complete news coverage. Explore the company directory ->

Source connectionsManage

Federal Reserve BoardPublic sourceLast fetch succeeded

European Central BankPublic sourceLast fetch succeeded

SEC EDGAR disclosuresPublic sourceLast fetch failed

BenzingaYour API connectionNeeds setup

MarketauxYour API connectionNeeds setup

FinnhubYour API connectionNeeds setup

Freshness & coverage

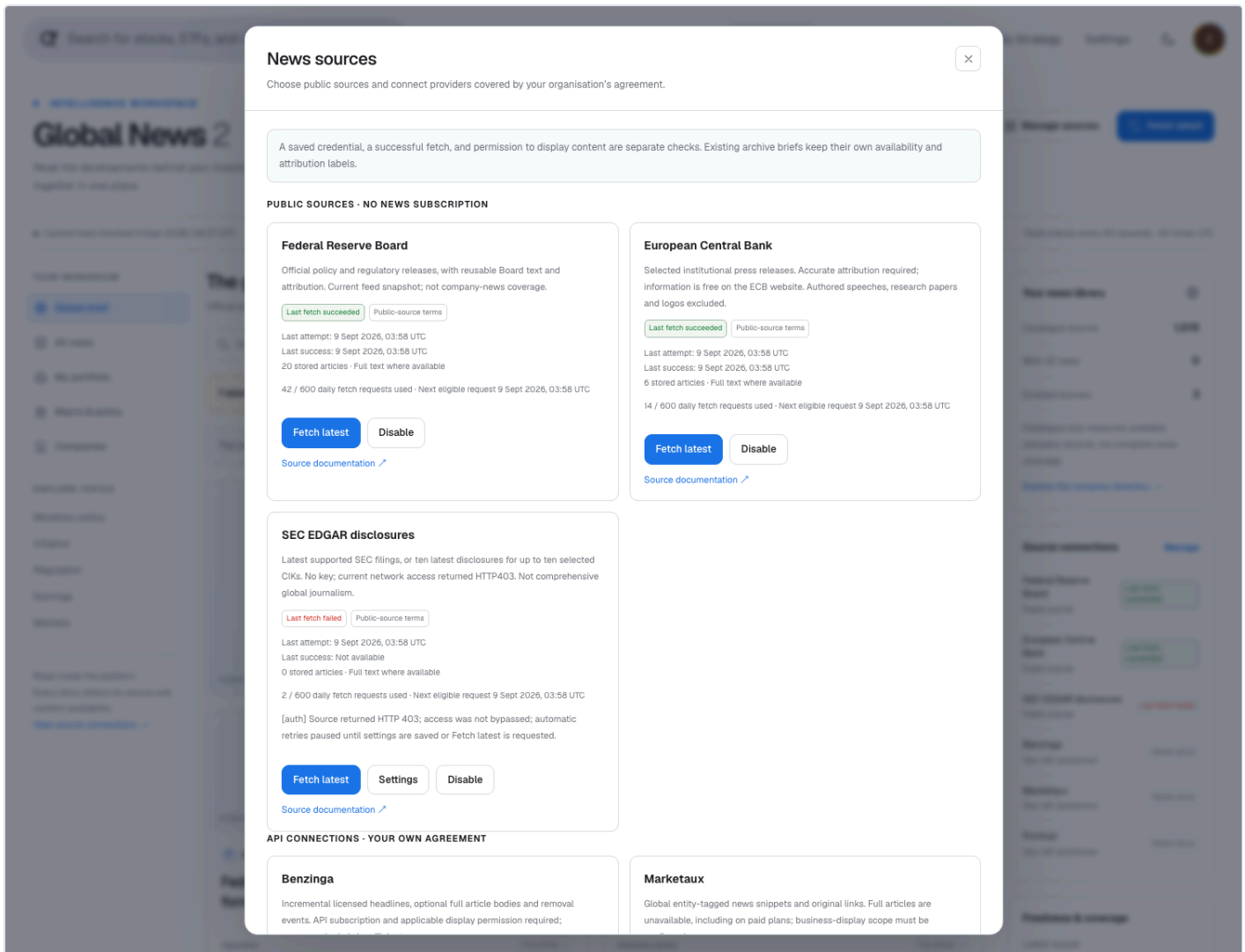
Latest receipt9 Sept 2026, 03:58 UTC

Stored coverage does not guarantee current news for every company. Official macro feeds are not company-wide coverage; consult each provider's success time and quota.

SEC EDGAR disclosuresfailed · 0 stored

European Central Bank succeeded · 0 stored

The 4 September Federal Reserve enforcement-release example contains 529 characters of retrieved release text. The reader preserves the source and original URL. It does not invent an article body from a headline.



Real connection settings: free sources plus optional API providers. No commercial key was saved or tested for this screenshot.

Search for stocks, ETFs, and more...

Company directory

1,515 company records in the news catalogue. Select a company to read its associated stories.

Search

Coverage varies by source and company. A catalogue listing does not mean recent news is available. Country and listing details identify the company; they do not filter the global feed.

A	Agilent Technologies Inc US · A	US
AAL	American Airlines Group US · AAL	US
AAMI	Acadian Asset Management Inc US · AAMI	US
AAON	AAON Inc US · AAON	US
AAPL	Apple Inc US · AAPL	US
ABBV	AbbVie Inc US · ABBV	US
ABNB	Airbnb Inc US · ABNB	US
ABT	Abbott Laboratories US · ABT	US
ABX	Abacus Global Management Inc US · ABX	US
ACA	Arcosa Inc US · ACA	US
ACAD	ACADIA Pharmaceuticals Inc US · ACAD	US
ACEL	Accel Entertainment Inc US · ACEL	US
ACGL	Arch Capital Group Ltd US · ACGL	BM

The news directory reads the 1,515-row service catalogue with pagination, separately from the original 50-company quote catalogue.

🔍 Search for stocks, ETFs, and more

My StocksNewsCatalogueGlobal NewsGlobal News 2PolymarketAlerts27My StrategySettings

INTELLIGENCE WORKSPACE

Global News 2

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Manage sourcesFetch latest

Cached feed checked 9 Sept 2026, 04:09 UTCFeed checks every 60 seconds · All times UTC

YOUR NEWSROOM

Global brief

All news

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Macro & policy

Companies

EXPLORE TOPICS

Monetary policy

Inflation

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Earnings

Markets

Read inside the platform. Every story retains its source and content availability. View source connections →

Your portfolio, in the news

Stories linked to companies matched to your holdings.

Search headlines and summariesSearchExisting archiveNo archive topic tags

3 matched companies from 3 holdings.

GOOGLASMLAAPL

Archive items are previously stored headlines, summaries and source links. Their publication dates and content permissions differ from current connected feeds.

1 source reports a failed last fetch. Available stories remain visible. Review source status

The latest available story in this selection was published 31 Aug 2026, 13:18 UTC. This feed reflects available source results, not a real-time news guarantee.

CNBC · 31 Aug

Broadcom reports earnings this week. Mike Khouw says this trade could be a winner

"The stakes are high" heading into Broadcom earnings, says Mike Khouw.

News brief →

BENZINGA · ILLUSTRATION

BENZINGA · ILLUSTRATION

BENZINGA · ILLUSTRATION

Your news library

Catalogue records1,515

With V2 news0

Enabled sources3

Catalogue size measures available company records, not complete news coverage. Explore the company directory →

Source connectionsManage

Federal Reserve Board

Public source

Last fetch succeeded

European Central Bank

Public source

Last fetch succeeded

SEC EDGAR disclosures

Public source

Last fetch failed

Benzinga

Your API connection

Needs setup

Marketaux

Your API connection

Needs setup

Finnhub

Your API connection

Needs setup

Freshness & coverage

Latest receipt

All three current holdings match. The view shows 18 real archived company stories; the newest was published on 31 August 2026. Archive status and the age warning remain visible. This is evidence of working company filters, not fresh free company coverage.

Operating the local version

Keep the existing V1 service on port 8080. Run the V2 service separately on localhost 8081 with `NEWS_V2_WORKER_ENABLED=true` and the V1 scheduler disabled. The root `.env` points `NEWS_V2_SERVICE_URL` to 8081. Both processes share the same `NEWS_V2_ADMIN_TOKEN`; the service additionally needs a random 32-byte base64 `NEWS_V2_ENCRYPTION_KEY`. Those values are server-only and have been configured in the local workspace without committing them.

The migration is additive. Paid providers remain disabled and unconfigured until an entitled user connects them. Default daily budgets are 600 actual HTTP requests per free source and 100 per paid source; these are conservative application caps, not purchased quota guarantees. Jobs and state survive restarts. Pausing/disconnecting a source prevents further display/fetch; credential and content cleanup must follow the exact provider agreement, including backup retention in a production system.

The V2 MVP proxy intentionally serves this single workspace only on localhost in development. Production requires authenticated tenant isolation before enabling it. No provider purchases, sales messages or public deployment were performed.

Sources

All external sources were checked on 9 September 2026. Repository findings come from code inspection and the local databases; provider access and pricing claims below are based on the providers' own documentation and terms.

1. Federal Reserve disclaimer and reuse policy; official RSS feeds.
2. ECB disclaimer, copyright, reuse and framing conditions; ECB RSS.
3. SEC EDGAR APIs and fair access.
4. SEC webmaster FAQ and public content; SEC RSS.
5. Benzinga API FAQ; newsfeed API guide.
6. Benzinga official AWS Marketplace product; product EULA.
7. Marketaux documentation; FAQ: snippets only.
8. Marketaux pricing.
9. Marketaux terms.
10. Finnhub terms: business use and redistribution; API documentation.
11. LSEG News overview; official news API pagination guidance.
12. Dow Jones official Factiva Analytics SDK.
13. ASML issuer share-listings table: NASDAQ ASML.
14. Alphabet Q2 2026 filing cover: GOOGL Class A and GOOG Class C.